

Eddi Xie

eddi.xie@mail.utoronto.ca | [GitHub](#) | [LinkedIn](#) | [portfolio](#)

EDUCATION

Master of Engineering (MEng) in Mechanical Engineering, Robotics Emphasis + Co-op

University of Toronto, Toronto, ON

Sep 2026 – Nov 2028 (expected)

- Relevant Courses: Perception for Robotics, State Estimation; Winter 2027: Motion Planning, Deep Learning, Nonlinear Control

Bachelor of Applied Science (BASc) in Mechanical Engineering + Data Science Minor

University of British Columbia, Vancouver, BC

May 2026

- Relevant Courses: Software Development for Mechatronic Systems (C++), Automatic Control, Modern Control Engineering, Applied Machine Learning, Digital Systems and Microcomputers, Statistical Modelling for Data Science

SKILLS

- **Programming:** C++17, Python, C, C#, TypeScript, SQL, MATLAB
- **Robot learning & perception:** LeRobot, imitation learning (ACT), policy ensembles, PyTorch, MuJoCo, inverse kinematics, OpenCV, YOLOv8, visual servoing, experimental design
- **Embedded & real-time:** 50 Hz control loops, state machines, fail-safe interlocks, Betaflight MSP, UART serial, TCP/UDP, Raspberry Pi, hardware-in-the-loop testing
- **Software & testing:** Git, GitHub Actions CI, CMake/CTest, pytest, Jest, vitest, FastAPI, Node.js, PostgreSQL, React, React Native, Claude API, UI/UX design
- **Hardware & design:** SO-101 assembly and calibration, soldering, 3D printing (PLA, TPU), SolidWorks, machining, G-code, requirements and AHP decision matrices, budgeting
- **Languages:** English, Dutch, Mandarin (conversational)
- Eligible for an 8-month co-op work-term from May 2027 to December 2027

ROBOTICS & ENGINEERING PROJECTS

Project Lead

Sep 2026 – Present

Two-Arm Imitation-Learning Rig, personal and Perception course project – Toronto, ON

- Built a leader–follower **SO-101** teleoperation rig in **LeRobot** for C\$2,264 of a C\$2,300 cap, printing both arms and compliant TPU fingers, then assembling and calibrating them.
- Leading an **active-vision** course project (in progress) aiming a camera by an **ACT** ensemble's disagreement; 2,880 **MuJoCo** viewpoints showed an SO-101 can be the camera arm.
- Trained a single policy for all camera strategies on 981 randomized-view episodes from 882 IK-solved, collision-checked poses, cutting its training from 35–88 h to 7.4–8.0 h.
- Ran a pre-registered 1,000-scene AV-ALOHA test: an uncertainty-steered camera beat a fixed one by **12 points** ($p < 10^{-9}$); a replayed-path control tested whether aim caused it.
- Extending it to occlusion (in progress): a third policy on the same episodes re-rendered with the rig camera's 90° lens and random occluders, where the view should matter more.

Flight Control Lead – [GitHub](#)

Sep 2025 – Sep 2026

Autonomous Object-Tracking Drone, course project then solo, UBC – Vancouver, BC

- Wrote the team's entire **C++** flight-control subsystem (3,493 of 3,496 lines): a **50 Hz** loop, a Betaflight MSP driver and a state machine with a kill latch and watchdog.
- Defined the vision–control–GUI interface contracts and Git workflow for a three-person, three-language team, and soldered the motor–ESC and UART links.

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- Rebuilt the stack solo after the course as four processes (**C++17**, Python, FastAPI, React) on a versioned JSON contract over UDP, TCP and WebSockets, for a Raspberry Pi.
- Built a **YOLOv8n** and **OpenCV** KCF/CSRT tracking pipeline and a proportional visual-servo law turning bounding-box error into yaw, forward and altitude commands.
- Built fail-closed interlocks, ~370 CI tests and a 16-item pre-flight gate; the first **hardware-in-the-loop** bench run still caught minimum throttle arriving as half throttle.

Technical Manager – nominated for top mech capstone Sep 2024 – Apr 2025
Imaging Phantom Printer, Capstone for QURIT (BC Cancer), UBC – Vancouver, BC

- Converted a paste-extrusion printer to **UV-curable resin** on a five-person team, enabling water-insoluble gradient phantoms; the client has since printed with radioisotopes.
- Owned the software and **G-code**, adapting a DICOM-to-STL pipeline for UV curing, cutting lamp passes per layer from 90 to 12 and dropping a 30-minute bed bake per print.
- Designed the dual-feed Y-nozzle in **SolidWorks**, prototyped it in 3D print, then **machined** it over ~30 shop hours after diagnosing stripped threads and low stiffness.
- Logged ~90 hours of printer and material testing and replaced resin-swollen Buna-N rotor seals with resin-resistant FKM, ending weeks of blocked printing.
- Co-owned 19 measurable requirements and AHP scoring; wrote the budget (23% under at delivery), the risk register, the operating procedure and the 60+ page final report.

EXPERIENCE

Co-Founder & Project Lead – [coddle.app](#) Aug 2025 – Present
Coddle, AI cooking assistant for iOS – Vancouver, BC

- Co-founded an AI cooking assistant shipped to a **TestFlight** beta, owning the backend and four of five app tabs (543 of 972 commits) and ~¼ of its 2,016 **Jest** tests.
- Designed the **PostgreSQL** schema (19 of 23 migrations, GIN full-text search, event-sourced log) and a 38-endpoint **Node.js** API with token auth and rate limits.
- Took the **Claude** recipe endpoint to production with a strict three-recipe contract, atomic writes and per-request token, latency and cost telemetry (\$0.018 per generation).
- Designed the Cook tab's UI/UX and the app's design system, and built the **React Native** cooking mode with wall-clock multi-timers and kill-safe session restore.
- Traced a beta tester's churn to a validation cap that failed 16 straight recipe requests while dashboards showed 100% success, and wrote the fix with 6 failing-first tests.

Engineering Intern, R&D and Project Lead May 2025 – Aug 2025
MYT Shutter – Zhuhai, China

- Designed a proof-of-concept **motorized shutter** (CAD, motor sizing, control electronics, RF remote) to test whether a drive unit fits inside the existing frame.
- Ran the R&D solo from concept through sourcing, assembly and tuning, working with suppliers in Mandarin and advising on a CNC machinery purchase.
- Delivered a prototype that passed functional testing and became the basis for production development with third-party manufacturers.